JINHAO LIANG

RESEARCH INTERESTS

I am particularly interested in integrating generative modeling with optimization to address complex engineering and scientific challenges. I am currently developing algorithms to ensure that the outputs of diffusion and flow-based generative models satisfy constraints with provable guarantees, with the goal of enabling scalable and adaptive solutions in robotics, biology, and physics.

EDUCATION

University of Virginia (UVA) — Sep. 2024 - May. 2029
Ph.D. in Computer Science — GPA: 3.77/4.00
Advisor: Prof. Ferdinando Fioretto

The Chinese University of Hong Kong, Shenzhen (CUHKSZ) — Sep. 2022 - Jul. 2024
M.Phil in Computer and Information Engineering — GPA: 3.90/4.00
Advisor: Prof. Chenye Wu

Xidian University (XDU) — Sep. 2018 - Jun. 2022
B.E. in Software Engineering — GPA: 3.70/4.00

EXPERIENCE

Argonne National Laboratory — Jun. 2025 - Aug. 2025
Research Intern
Mentor: Sandeep Madireddy

University of Virginia — Sep. 2024 - Now
Graduate Research Assistant
Mentor: Ferdinando Fioretto

PUBLICATIONS

Machine Learning Conference Publications

1. Prince Zizhuang Wang, Shuyi Chen, **Jinhao Liang**, Ferdinando Fioretto, and Shixiang Zhu, "Gen-dfl: Decision-focused generative learning for robust decision making." accepted by ***The Fourteenth International Conference on Learning Representations (ICLR)***, 2026
   IISE DAIS Best Student Paper Finalist.
2. **Jinhao Liang**, Yixuan Sun, Anirban Samaddar, Sandeep Madireddy and Ferdinando Fioretto, "Chance-constrained Flow Matching for High-Fidelity Constraint-aware Generation." accepted by ***NeurIPS 2025 MLXOR Workshop and NeurIPS 2025 COML Workshop***, 2025
3. **Jinhao Liang**, Sven Koenig and Ferdinando Fioretto, "Discrete-Guided Diffusion for Scalable and Safe Multi-Robot Motion Planning." accepted by ***The 40th Annual AAAI Conference on Artificial Intelligence (AAAI)***, 2025
4. Jacob K. Christopher, Michael Cardei, **Jinhao Liang**, and Ferdinando Fioretto, "Neuro-Symbolic Generative Diffusion Models for Physically Grounded, Robust, and Safe Generation." accepted by ***The 2nd International Conference on Neuro-symbolic Systems (NeuS)***, 2025.
5. **Jinhao Liang**, Jacob K. Christopher, Sven Koenig and Ferdinando Fioretto, "Simultaneous Multi-Robot Motion Planning with Projected Diffusion Models." accepted by ***The 42nd International Conference on Machine Learning (ICML)***, 2025.

6. **Jinhao Liang**, Jacob K. Christopher, Sven Koenig and Ferdinando Fioretto, "Multi-Agent Path Finding in Continuous Spaces with Projected Diffusion Models." accepted by ***The 6th International Workshop on Multi-Agent Path Finding, at AAAI***, 2025.

Journal Articles

1. **Jinhao Liang**, Wenqian Jiang, Chenbei Lu and Chenye Wu, "Joint Chance-constrained Unit Commitment: Statistically Feasible Robust Optimization with Learning-to-Optimize Acceleration." accepted by ***IEEE Transactions on Power Systems***, 2024. (JCR Q1)
2. Chenbei Lu, **Jinhao Liang**, Nan Gu, Haoxiang Wang and Chenye Wu, "Manipulation-Proof Virtual Bidding Mechanism Design." accepted by ***IEEE Transactions on Energy Markets, Policy and Regulation***, 2023.
3. Chenbei Lu, **Jinhao Liang**, Wenqian Jiang, Jiaye Teng and Chenye Wu, "High-Resolution Probabilistic Load Forecasting: A Learning Ensemble Approach." accepted by ***Journal of the Franklin Institute***, 2023. (JCR Q1)

Power & Control Conference Publications

1. Chenbei Lu, **Jinhao Liang**, Hongyu Yi and Chenye Wu, "Cost-effective Closed-loop Bilevel Robust Optimization for Joint Chance-constrained Economic Dispatch." accepted by ***The 16th ACM International Conference on Future and Sustainable Energy Systems (e-Energy)***, 2025.
2. **Jinhao Liang**, Ferdinando Fioretto and Chenye Wu, "Robust Bidding Strategies in Local Energy Markets." accepted by ***2025 IEEE Power & Energy Society General Meeting (PESGM)***, 2025
3. **Jinhao Liang**, Chenbei Lu, Wenqian Jiang and Chenye Wu, "Few-shot Residential Load Forecasting Boosted by Learning to Ensemble." accepted by ***the 7th IEEE Conference on Energy Internet and Energy System Integration (EI^2)*** , 2023
4. Wenqian Jiang, **Jinhao Liang**, Chenbei Lu and Chenye Wu, "Robust Online EV Charging Scheduling with Statistical Feasibility." accepted by ***the 62nd IEEE Conference on Decision and Control (CDC)***, 2023
5. **Jinhao Liang**, Wenqian Jiang and Chenye Wu, "Effective Carbon Tax Learning via Cap and Trade." accepted by ***IEEE 5th International Electrical and Energy Conference (CIEEC)***, 2022

SELECTED RESEARCH EXPERIENCE

Chance-constrained Flow Matching for High-Fidelity Constraint-aware Generation
Supervisor: Prof. Ferdinando Fioretto and Sandeep Madireddy — Feb. 2025 - Jul. 2025

- Integrated stochastic optimization into flow-matching generative models, enabling **concise and effective enforcement of hard constraints while preserving the high-fidelity sample generation**.
- Established a theoretical framework for enforcing hard constraints in the sampling process of generative models, demonstrating the **feasibility guarantees and sample quality preservation theoretically**.
- Achieved **2.3× higher feasibility** and over **20× higher efficiency** than prior constrained generative models.

Discrete-guided Diffusion for Scalable and Safe Multi-Robot Motion Planning
Supervisor: Prof. Ferdinando Fioretto — Feb. 2025 - Jul. 2025

- Decomposed the original MRMP problem into a sequence of local subproblems with convex configuration spaces.
- Guided diffusion models with spatiotemporal structure derived from the MAPF solution to generate high-quality trajectories.
- Designed a lightweight projection mechanism that enforces feasibility and activates only when constraint violations are detected.

Multi-Robot Motion Planning with Projected Diffusion Models
Supervisor: Prof. Ferdinando Fioretto Sep. 2024 - Jan. 2025

- Introduced a novel formulation of Multi-Robot Motion Planning (MRMP) in continuous spaces using diffusion models, enabling the simultaneous generation of trajectories for all agents in a single framework.
- Adapted projected diffusion models (PDM) for MRMP by embedding constraints directly into the diffusion process, ensuring that the generated solutions are feasible and collision-free.
- Developed an augmented Lagrangian approach to accelerate the projection process, making the method scalable and practical for real-world applications.

Joint Chance-constrained Unit Commitment with Statistical Feasibility
Supervisor: Prof. Chenye Wu Oct. 2022 - Jun. 2023

- Extended the notion of statistical feasibility into unit commitment, a mixed-integer problem, and formulated the statistically feasible unit commitment.
- Developed sample-based uncertainty set construction algorithms, yielding less conservative solutions.
- Accelerated the solving process and designed the optimization kernel to boost its computational efficiency further.

SELECTED TALKS

"Constrained Flow Matching for Scientific Applications" Mar. 2026
Minisymposium on *Application of Flow-based Generative Models in Science*, SIAM UQ26.

ACADEMIC SERVICES

Reviewer

- *IEEE Transactions on Power Systems*
- *Applied Energy*
- *IEEE Transactions on Energy Markets, Policy, and Regulation*
- *IEEE Transactions on Emerging Topics in Computational Intelligence*
- *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*
- *AAAI Conference on Artificial Intelligence (AAAI)*
- *International Conference on Learning Representations (ICLR)*

SELECTED HONORS

ICML 2026 Silver Reviewer Recognition	May 2026
FPI AITHYRA Travel Grant	Nov 2025
DARPA Disruptive Idea Award	May 2025
WoMAPF student travel grants	Dec 2024
AAAI Student Scholarship	Dec 2024
Presidential Award for Outstanding Graduate Students (top 10 in CUHKSZ)	Nov 2024
Provost Fellowship (top graduate students in UVA)	Sep 2024